PART A: OBJECTIVES

Answer ALL questions in answer booklet.

1. What is a multiplexer?
   A. It is a type of decoder which decodes several inputs and gives one output.
   B. A multiplexer is a device that has many inputs and a single output.
   C. It takes one input and results into many outputs.
   D. It is a type of encoder which decodes several inputs and gives one output.

4. In a multiplexer the output depends on its _____.
   A. data outputs
   B. select inputs
   C. select outputs
   D. rotate pin

5. Parity generators/checkers use which modulo summation?

   A. Modulo 8

   B. Modulo 16

   C. Modulo 2

   D. Modulo 10

6. A code converter is a logic circuit that _____.

   A. Inverts the given input

   B. Converts into decimal number

   C. Converts data of one type into another type

   D. Converts to hexadecimal

PART B: STRUCTURED
Answer ALL questions in question and answer booklet.

Question 1 [16 Marks]

a) Draw the block diagram for a 3-bit parallel adder using full adder and label clearly the block diagram including the input, sum and carry. Perform addition $011_2 + 110_2$ **,** show the value for each label on the block diagram. [4]

$$\begin{array}{rcccc} & A_3 & A_2 & A_1 \\ + & B_3 & B_2 & B_1 \\ \hline \sum_4 & \sum_3 & \sum_2 & \sum_1 \end{array}$$

b) For set of binary numbers $A_3A_2A_1A_0 = 0010$ $B_3B_2B_1B_0 = 0101$ are assigned to comparator in Figure 1.

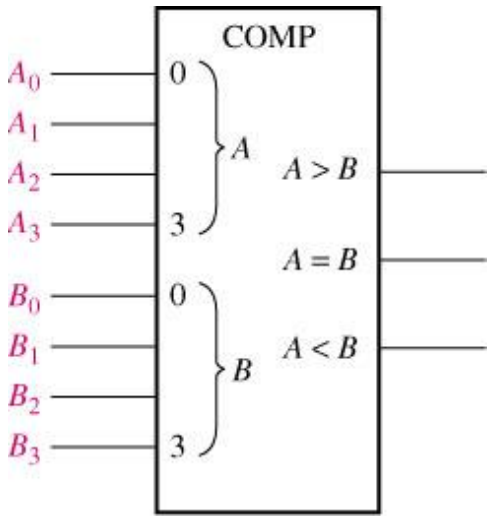


Figure 1

i. Based on Figure 1, explain how comparator works until final output results appear. [3]

ii. Determine the final output result for the comparator in Figure 1. [1]

c) Given an active HIGH 7-segment display in Figure 2 showing a digital number from a decoder that is based on the BCD value, **complete Table 1 in your question booklet APPENDIX A**. [4]

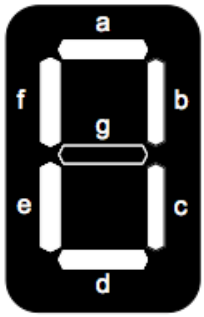


Figure 2

d) Figure 3 in APPENDIX A shows DEMUX connection with outputs labelled as Y_3 Y_2 Y_1 Y_0 and B is the input of DEMUX. Select bit for DEMUX are noted by C_1 and C_0 (C_1 is the MSB). **Draw the waveform for the DEMUX output based on the given values in question booklet Figure 3 APPENDIX A.** [4]